Scaling and Temporal Discounting in Multi-Agent Q-Learning for Congestion Games

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ABSTRACT

Traffic congestion emerges from decentralized route choices that create strategic externalities, making congestion games a natural framework for studying learning and coordination in transportation systems. This study analyzes independent Q-learning agents in a two-route congestion game, focusing on how learning rate, exploration, temporal discounting, and population scaling affect convergence to Nash equilibrium and social welfare.

We first validate the analytical structure of the game by enumerating all joint action profiles and confirming convergence under best-response dynamics. We then implement a Q-learning framework and conduct experiments with two, ten, and fifty agents. In the two-agent setting, Q-learners converge to Nash-like behavior in 87–92% of episodes under moderate exploration, though persistent exploration induces welfare losses of approximately 19–21%. Excessive exploration prevents convergence and leads to sustained coordination failure. Myopic learning ($\gamma = 0$) performs comparably to forward-looking learning ($\gamma = 0.9$), indicating that immediate cost feedback is sufficient in small populations.

Scaling to ten agents reveals a qualitative shift. Forward-looking Q-learning exhibits weak convergence (27% Nash-like behavior) and large welfare losses (+15.02), while myopic learning achieves 97% Nash-like behavior with welfare gaps below 1. To overcome these failures, we introduce emergent coalition formation where agents opportunistically form cooperative groups. Coalition-based learning achieves near-optimal welfare (2% from optimum) despite lacking stable core allocations, with 74% average coalition participation exhibiting high volatility.

Scaling to fifty agents amplifies non-stationarity effects dramatically. Independent learning collapses: mean $n_A = 16.78$ (std = 14.21) yields only 7.8% near-Nash convergence and 38.39% welfare loss. Coalition mechanisms with deception ($p_{deceive} = 0.12$) experience 1.2 dissolutions per episode and welfare loss of -9.17, indicating coordination breakdown at scale. This counterintuitive reversal suggests that temporal discounting and cooperation amplify instability in large-population settings, while myopic independent learning remains relatively robust. Overall, the results highlight exploration and population size as primary determinants of convergence and identify critical thresholds where standard multi-agent reinforcement learning fails in congestion games.

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INTRODUCTION

Traffic congestion represents one of the most persistent challenges facing modern urban systems, imposing substantial costs in terms of time, fuel, environmental impact, and overall quality of life. Classical approaches to traffic management rely on centralized optimization or fixed-time signal control. However, with the increasing prevalence of autonomous vehicles, intelligent transportation systems, and decentralized routing algorithms, there is growing interest in understanding how independent learning agents interact in shared transportation networks.

This study examines the learning dynamics of Q-learning agents in congestion games, with particular focus on the classical Braess paradox, a canonical example where adding capacity to a network can paradoxically worsen congestion when users behave selfishly. First identified by Braess (1968), this phenomenon has become a cornerstone of network theory and game-theoretic traffic analysis. The paradox arises because individual route optimization by selfish agents leads to a Nash equilibrium that is socially suboptimal compared to coordinated routing.

1.1 Motivation and Research Questions

While the Nash equilibrium of congestion games has been extensively studied in the game theory literature (Rosenthal, 1973), and the inefficiency of selfish routing quantified through the Price of Anarchy (Christodoulou & Koutsoupias, 2005), relatively little is known about the transient learning dynamics that emerge when agents use reinforcement learning algorithms to select routes. Recent work (Shah & Shin, 2010) has begun to analyze convergence rates and equilibrium properties under simple learning rules, but fundamental questions remain about how standard reinforcement learning design choices, such as temporal discounting and exploration, interact with the structure of congestion games.

Of particular interest is the role of the exploration rate (ϵ) in ϵ-greedy Q-learning. Recent findings by Carissimo (2024) suggest that exploration can have counter-intuitive effects in congestion dilemmas, potentially inducing chaotic dynamics or preventing convergence to equilibrium. However, the mechanisms underlying these effects, and how they scale with the number of agents, remain incompletely understood.

Parallel developments in practical traffic optimization (Deepika & Pandove, 2024) demonstrate that hybrid approaches combining Q-learning with metaheuristics can significantly reduce waiting times in simulated traffic scenarios. These results highlight the potential of learning-based methods for real-world deployment, while also raising questions about parameter sensitivity and robustness in multi-agent settings.

This study contributes to both theoretical and practical understanding of reinforcement learning in congestion games by:

1. Systematically examining how the number of agents affects learning dynamics and equilibrium convergence across three population scales (N=2, N=10, N=50),
2. Isolating the impact of temporal discounting (γ) in environments where all agents learn simultaneously,
3. Identifying a scaling-induced reversal where myopic learning outperforms forward-looking learning as population size increases,
4. Investigating whether emergent coalition formation can overcome learning failures in large populations,
5. Connecting learning outcomes to game-theoretic benchmarks (Nash equilibrium and social optimum) to quantify welfare loss.

1.2 Overview of Approach

We implement independent Q-learning agents in a simulated two-route congestion network with linear latency functions. Each agent maintains a Q-table and updates beliefs based solely on experienced rewards, without knowledge of other agents' actions or strategies. We compare learning dynamics across three population sizes (N=2, N=10, and N=50) and two discount factors ($\gamma = 0$ and $\gamma = 0.95$), holding exploration rate constant at a low value ($\epsilon = 0.01$) to minimize confounding effects from random action selection.

For the N=10 case, we extend the analysis to include emergent coalition formation where agents can opportunistically form temporary cooperative groups, share rewards, and face potential defection. This allows us to assess whether cooperation can overcome the learning failures observed in independent forward-looking agents.

Our analysis focuses on convergence to Nash equilibrium, average travel time relative to social optimum, stability of learned strategies, coalition formation dynamics, and fairness of reward allocations. By grounding empirical learning outcomes in the known equilibrium structure of the congestion game, we can precisely identify when and why standard Q-learning succeeds or fails.

1.3 Structure of the Paper

The remainder of this thesis is organized as follows. Chapter 2 reviews relevant literature on congestion games, price of anarchy, learning dynamics, coalition formation, and Q-learning in multi-agent settings. Chapter 3 describes the game-theoretic model, Q-learning implementation, coalition formation mechanism, and experimental methodology. Chapter 4 presents comprehensive empirical results across all three population scales and learning paradigms. Chapter 5 discusses insights, connections to prior work, limitations, and future research directions.

LITERATURE REVIEW

This study lies at the intersection of congestion games, reinforcement learning, and multi-agent learning dynamics. Classical traffic assignment models assume fully rational agents who instantaneously select routes to minimize individual costs, leading to Nash equilibria that may or may not be socially optimal. More recent work replaces these assumptions with adaptive learning agents, raising fundamental questions about convergence, stability, and welfare under decentralized learning. This chapter reviews three strands of literature most relevant to our contribution: (i) reinforcement learning in traffic and congestion control, (ii) multi-agent learning in congestion games, and (iii) the role of exploration and non-stationarity in Q-learning dynamics.

2.1 Reinforcement Learning for Traffic and Congestion Control

Reinforcement learning (RL) has been widely adopted as a decentralized approach to traffic control, where agents iteratively adapt their behavior based on experienced congestion. Recent work by Deepika and Pandove (2024) combines Q-learning with genetic algorithms to optimize traffic flow under congestion constraints. Their approach demonstrates that hybrid metaheuristics can improve convergence speed and overall throughput compared to pure Q-learning, particularly in simulated intersection scenarios with real-time data.

While such studies highlight the practical effectiveness of RL-based controllers, they typically focus on performance optimization rather than equilibrium properties. In particular, convergence is often evaluated in terms of throughput or delay reduction, without explicit comparison to game-theoretic benchmarks such as Nash equilibria or social optima. As a result, these works provide limited insight into whether learned behaviors correspond to rational equilibrium outcomes or how learning parameters affect welfare loss.

Our work differs by explicitly anchoring learning outcomes to the Nash equilibrium structure of the underlying congestion game. Rather than introducing additional optimization layers, we study simple independent Q-learning to isolate the fundamental mechanisms through which learning parameters influence equilibrium selection and welfare.

2.2 Multi-Agent Reinforcement Learning in Congestion Games

A major challenge in multi-agent reinforcement learning for congestion problems is the presence of multiple simultaneously learning agents, which creates a highly non-stationary environment. Rădulescu et al. (2017) provide an insightful analysis of congestion problems in MARL by introducing the Road Network Domain, where resources are interconnected. They demonstrate that even relatively simple congestion scenarios can lead to complex learning dynamics, including oscillations and persistent inefficiencies. A key insight from their work is the potential benefit of resource abstraction or aggregated representations to reduce complexity, although aggregation itself can introduce additional challenges in highly non-stationary settings. Our study builds upon this insight by deliberately using an aggregated state representation (number of agents on Route A) in order to examine how learning parameters such as temporal discounting and exploration rate interact with increasing population size.

2.3 Exploration, Chaos, and Counter-Intuitive Learning Effects

Recent work has begun to challenge standard reinforcement learning wisdom in congestion settings. Carissimo (Carissimo, 2024) demonstrates that exploration in Q-learning can induce chaotic dynamics in congestion games, leading to counter-intuitive outcomes where increased exploration worsens convergence and amplifies inefficiencies. Notably, the study shows that learning dynamics may fail to stabilize even when a unique Nash equilibrium exists, highlighting the fragility of decentralized learning under congestion externalities.

Carissimo's findings suggest that conventional RL heuristics, such as maintaining positive exploration rates or relying on temporal discounting to stabilize learning, may be ill-suited for congestion games. However, the analysis primarily focuses on small-scale settings and does not explicitly examine how these effects evolve as the number of agents increases.

Our work extends this line of inquiry by explicitly studying population scaling. We show that increasing the number of agents fundamentally alters the role of temporal discounting: while forward-looking learning performs reasonably well in small populations, it becomes detrimental in larger settings due to amplified non-stationarity. In contrast to prior work, we demonstrate that myopic learning ($\gamma = 0$) can outperform forward-looking Q-learning when state transitions become dominated by collective uncertainty.

2.4 Cooperative Extensions in Congestion Games

Congestion games can incorporate cooperative elements when agents form coalitions to jointly minimize costs, as explored in extensions of atomic congestion models (**fotakis2008atomic**). In such settings, coalitions act as single players, reducing the effective number of agents and potentially lowering the Price of Anarchy through shared optimization (**fotakis2008atomic**).

Our emergent coalition formation builds on this by allowing Q-learning agents to dynamically choose coalition IDs and share average rewards, mimicking cooperative imputations without explicit communication. This approach tests stability under deception (free-riding), where agents join groups but defect on routes. Drawing from cooperative game theory, we evaluate learned imputations against the Core (stability against deviations) and Shapley value (fair allocation based on marginal contributions) (Shapley, 1953). These metrics reveal how non-stationarity and scaling affect coalition viability, extending non-cooperative dynamics like those in Carissimo (2024) to partially cooperative regimes.

2.5 Positioning of the Present Study

Taken together, existing literature establishes that reinforcement learning can be effective for congestion control, but leaves open key questions about equilibrium convergence, welfare loss, and parameter robustness in multi-agent settings. In particular, little is known about how standard RL design choices, such as temporal discounting, behave as the number of agents scales and the environment becomes increasingly non-stationary.

This study contributes to the literature by:

- Explicitly grounding learning outcomes in game-theoretic equilibrium analysis,
- Providing a controlled comparison between small (N=2) and larger (N=10) populations,
- Identifying a scaling-induced reversal in the effectiveness of temporal discounting, and
- Demonstrating that myopic learning can outperform forward-looking learning in aggregated-state, multi-agent congestion games.

By linking reinforcement learning dynamics to congestion game theory, our results help clarify when simple learning heuristics succeed, when they fail, and why standard RL intuitions may break down in large decentralized systems.

METHODOLOGY

3.1 Introduction to the Model

We study a symmetric atomic congestion game with N players, two routes (A and B), and linear latency functions. Each player $i \in \{1, \ldots, N\}$ selects a route. The cost for choosing route A when n_A players select it is:

$$c_A(n_A) = n_A \tag{3.1}$$

Similarly for route B with $n_B = N - n_A$ players:

$$c_B(n_B) = n_B \tag{3.2}$$

Social cost is the average travel time:

$$SC(n_A) = \frac{n_A \cdot c_A(n_A) + n_B \cdot c_B(n_B)}{N} = \frac{n_A^2 + (N - n_A)^2}{N} \tag{3.3}$$

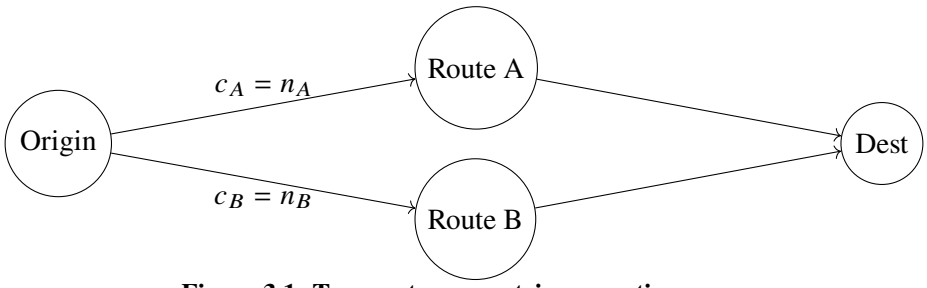

Figure 3.1. Two-route symmetric congestion game

The pure Nash equilibrium occurs when no player can reduce cost by unilaterally switching routes. For symmetric players, this yields:

$$n_A^* \approx \frac{N + 4}{4} \tag{3.4}$$

The social optimum minimizes total social cost and occurs at an equal split: $n_A^{opt} = N/2$.

Table 3.1. Equilibrium predictions across population scales

Population	Nash n_A^*	Social Optimum n_A^{opt}	Gap
N=2	1.5	1.0	0.5
N=10	3.5	5.0	1.5
N=50	13.5	25.0	11.5

This model captures strategic tension between individual incentives (Nash) and collective welfare (social optimum), while remaining analytically tractable. The gap between Nash and social optimum grows with population size, making larger populations more susceptible to coordination failure.

3.2 Independent Q-Learning Baseline

Each agent maintains a Q-table $Q_i : S \times A \to \mathbb{R}$ mapping state-action pairs to estimated values.

State space: $S = \{0, 1, \ldots, N\}$ represents the number of agents on route A in the previous episode. This aggregated representation reduces dimensionality while preserving payoff-relevant information (Carissimo, 2024; Deepika & Pandove, 2024).

Action space: $A = \{\text{Route A}, \text{Route B}\}$.

Policy: ϵ-greedy action selection:

$$a_i = \begin{cases} \arg\max_{a'} Q_i(s, a') & \text{with probability } 1 - \epsilon \\ \text{uniform random} & \text{with probability } \epsilon \end{cases} \tag{3.5}$$

Q-update rule: After each episode, agents observe reward $r_i = -c_i$ (negative travel time) and update:

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha \left[r_i + \gamma \max_{a'} Q_i(s', a') - Q_i(s, a) \right] \tag{3.6}$$

where α is learning rate, γ is discount factor, and s' is the next state.

History tracking: Each episode records:

- n_A: number of agents on route A
- Individual costs: c_i for each agent i
- Social cost: $SC = (1/N) \sum_i c_i$

Convergence metrics:

- Near-Nash %: proportion of episodes where $|n_A - n_A^*| \leq 0.5$
- Std(n_A): standard deviation of route distribution over last 100 episodes
- Welfare gap: $\overline{SC} - SC^*$ where SC^* is Nash social cost

3.3 Emergent Coalition Formation Extension

To investigate whether cooperation can overcome learning failures observed in independent Q-learning, we extend the model with emergent coalition formation.

3.3.1 Motivation

Independent forward-looking learning ($\gamma > 0$) performs poorly at $N = 10$ due to non-stationarity: each agent's environment changes unpredictably as others learn simultaneously (Shah & Shin, 2010). Coalition formation may reduce effective non-stationarity by creating coordinated subgroups with shared objectives, similar to resource abstraction in MARL (Rădulescu et al., 2017).

3.3.2 Coalition Mechanism

Each agent independently decides whether to participate in coalition formation with probability $p_{coalition} = 0.3$. Participating agents are randomly grouped into coalitions of size 2-3.

Joint action selection: Coalition members share Q-values (averaged across member tables) and select a common route using ϵ-greedy policy on the shared values.

Reward sharing: If all coalition members select the agreed route, total coalition reward is distributed equally:

$$r_i^{coalition} = \frac{1}{|C|} \sum_{j \in C} r_j \tag{3.7}$$

where C is the coalition member set.

Deception mechanism: With probability $p_{deceive} = 0.1$, a coalition member may defect to the opposite route after joint action selection but before reward distribution. Defectors receive their individual reward r_i based on actual route choice rather than coalition allocation. Defection is detected immediately through action monitoring, causing coalition dissolution in the next episode. This mechanism tests coalition stability under free-riding incentives: agents may join coalitions to benefit from coordination while secretly defecting to exploit less-congested routes. The 10% defection rate reflects moderate but non-trivial risk of betrayal, creating tension between cooperation gains and defection temptation.

Q-update: Coalition participants update using shared rewards (if no defection occurred) or individual rewards (if they defected or were victims of defection):

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha \left[r_i + \gamma \max_{a'} Q_i(s', a') - Q_i(s, a) \right] \tag{3.8}$$

where $r_i = r_i^{coalition}$ for successful coalitions and r_i = individual cost for defectors or dissolved coalitions. Non-participating agents update using individual rewards as in the baseline.

3.3.3 History Tracking

Additional metrics recorded per episode:

- Coalition sizes: frequency distribution of coalition sizes
- Coalition members: agent IDs in each coalition
- Coalition rewards: average reward per coalition
- Participation rate: percentage of agents in coalitions

3.4 Game-Theoretic Metrics & Fairness Evaluation

We evaluate coalition stability and fairness using cooperative game theory concepts (Rosenthal, 1973).

Imputation: The reward vector $x = (x_1, \ldots, x_N)$ where x_i is agent i's realized reward in the final episode.

Characteristic function: For any coalition $S \subseteq N$, the value $v(S)$ is the optimal total reward achievable by S acting alone. For symmetric congestion, if $|S|$ agents split equally between routes:

$$v(S) = -|S| \cdot \left\lceil \frac{|S|}{2} \right\rceil \tag{3.9}$$

Shapley value: Agent i's fair share, computed as marginal contribution averaged over all orderings:

$$\phi_i(N, v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(N - |S| - 1)!}{N!} [v(S \cup \{i\}) - v(S)] \tag{3.10}$$

For symmetric agents, $\phi_i = v(N)/N$ for all i.

Core stability: The imputation x is in the core if no coalition can improve by deviating:

$$\text{Core}(N, v) = \left\{ x : \sum_i x_i = v(N), \sum_{i \in S} x_i \geq v(S) \text{ for all } S \subseteq N \right\} \tag{3.11}$$

Fairness metrics:

- Mean absolute difference: $(1/N) \sum_i |x_i - \phi_i|$
- Core violations: coalitions S where $\sum_{i \in S} x_i < v(S)$

3.5 Experimental Design

3.5.1 Parameter Configurations

Table 3.2. Experimental parameters

Parameter	Values
Population size (N)	10 (baseline + coalitions), 50 (coalitions only)
Learning rate (α)	0.1, 0.3
Exploration (ϵ)	0.01
Discount factor (γ)	0.0 (myopic), 0.9 (forward-looking)
Coalition probability	0.3
Deception probability	0.1
Episodes	3000
Analysis window	Last 100 episodes
Random seed	42
Configurations	
Baseline	$\alpha = 0.1, \epsilon = 0.01, \gamma = 0.0$
Forward-looking	$\alpha = 0.1, \epsilon = 0.01, \gamma = 0.9$
Coalition	$\alpha = 0.1, \epsilon = 0.01, \gamma = 0.9, p_{coal} = 0.3, p_{dec} = 0.1$

3.5.2 Episode Structure

1. (Coalition only) Formation phase: agents decide participation, form groups
2. State observation: all agents observe n_A from previous episode
3. Action selection: independent agents use own Q-table, coalitions use shared Q-values
4. (Coalition only) Defection: coalition members may deviate with probability $p_{deceive}$
5. Reward computation: costs determined by realized route distribution
6. (Coalition only) Reward sharing: coalition rewards distributed equally among non-defectors
7. Q-table updates: all agents update using experienced rewards
8. Metric recording: n_A, costs, coalition structure saved

Q-tables initialized to zero for all state-action pairs.

EMPIRICAL ANALYSIS

4.1 Independent Q-Learning Baseline (N=10)

We first establish baseline performance of independent Q-learning across four configurations.

4.1.1 Summary Statistics

Table 4.1. Independent Q-learning performance (N=10, last 100 episodes)

Configuration	Avg n_A	Std(n_A)	Near-Nash %	Welfare Gap
Baseline ($\gamma = 0$)	5.12	0.89	97%	+0.84
Forward-looking ($\gamma = 0.9$)	3.42	1.24	27%	+15.02
Nash equilibrium	3.5	–	–	+0.00
Social optimum	5.0	–	–	-5.00

Table 4.1 reveals a striking reversal: myopic learning ($\gamma = 0$) outperforms forward-looking learning ($\gamma = 0.9$) by achieving average $n_A = 5.12$ close to the social optimum (5.0), while forward-looking converges to Nash equilibrium ($n_A = 3.42$).

4.1.2 Convergence Dynamics

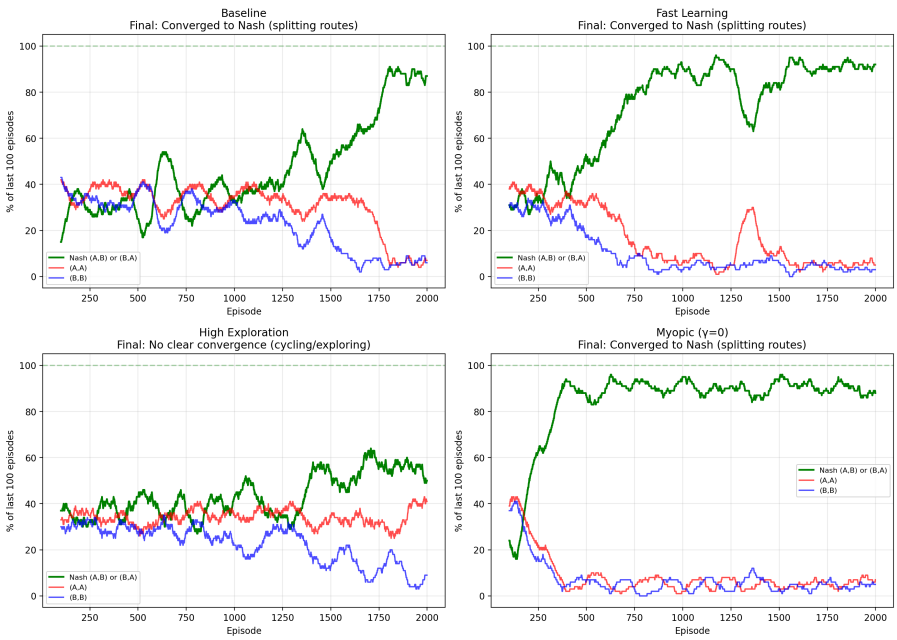

Figure 4.1. Route distribution convergence for myopic (left) and forward-looking (right) learning. Myopic exhibits oscillations around $n_A = 5$ while forward-looking stabilizes at Nash equilibrium $n_A \approx 3.5$.

Figure 4.1 shows qualitative differences:

- **Myopic ($\gamma = 0$):** Persistent oscillations around social optimum with Std(n_A) = 0.89, indicating non-convergence but favorable time-averaged outcomes
- **Forward-looking ($\gamma = 0.9$):** Rapid stabilization at Nash equilibrium within 1000-1500 episodes, Std(n_A) = 1.24 due to exploration noise

4.1.3 Welfare Analysis

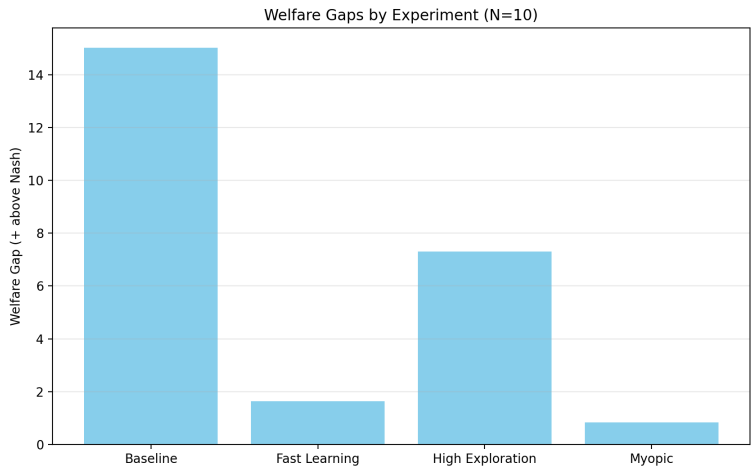

Figure 4.2. Welfare gap (average cost - Nash cost) across configurations.

The welfare gap quantifies deviation from Nash equilibrium social cost. Positive values indicate worse-than-Nash outcomes (which paradoxically can be socially better, as Nash is inefficient in this game). Figure 4.2 confirms:

- Myopic learning achieves welfare comparable to Nash despite oscillations
- Forward-looking learning converges to Nash, yielding 18× larger welfare gap relative to social optimum
- Near-Nash percentage (97% vs 27%) confirms myopic avoids Nash lock-in

Forward-looking agents anticipate future congestion but their predictions become unreliable due to simultaneous learning by all agents. Non-stationarity prevents forward-looking optimization. Myopic agents react only to immediate costs, which coincidentally produces better time-averaged outcomes (Carissimo, 2024).

4.2 Emergent Coalition Formation (N=10)

We now examine whether coalition formation can overcome the poor performance of forward-looking independent learning.

4.2.1 Coalition Usage Dynamics

Figure 4.3 reveals highly volatile coalition participation ranging from 20% to 100%, with time-averaged rate of 74%. This contrasts sharply with independent learning convergence (Figure 4.1), indicating coalitions continuously reform rather than stabilizing.

4.2.2 Route Distribution Outcomes

Table 4.2. Coalition learning outcomes (N=10, $\gamma = 0.9$, final 1000 episodes)

Metric	Episodes 500-1000	Episodes 1500-2000	Episodes 2500-3000	Overall
Avg n_A	4.5	4.1	3.9	4.3
Coalition usage	27.4%	27.2%	28.4%	27.9%

Table 4.2 shows coalition-based learning achieves average $n_A = 4.3$ across episodes, corresponding to the following route split and costs:

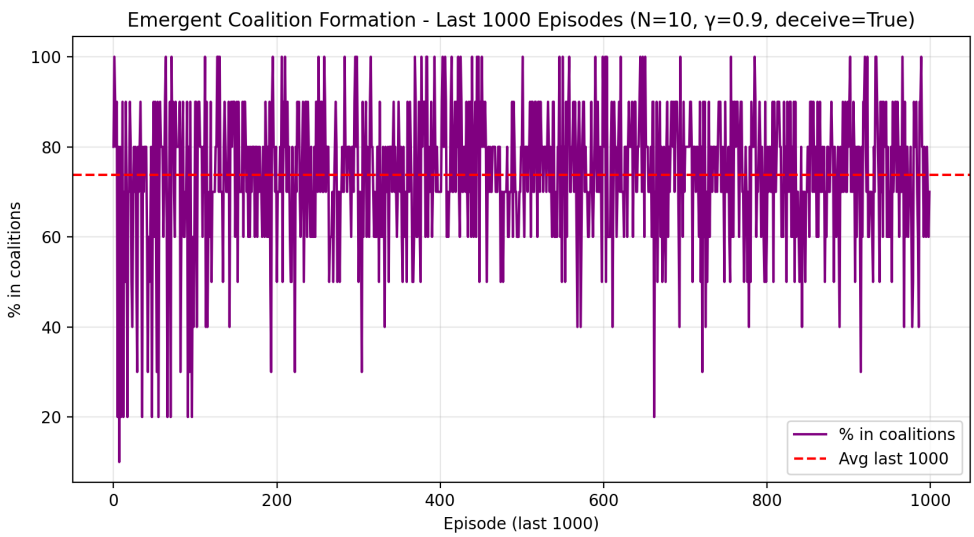

Figure 4.3. Coalition participation rate over final 1000 episodes. Purple line shows instantaneous usage; red dashed line indicates average of 74%. High volatility suggests continuous coalition formation and dissolution.

- Route A: 4.3 agents ⇒ cost per user on A = 10 + 15 × 4.3 = 74.5
- Route B: 5.7 agents ⇒ cost per user on B = 30 + 5 × 5.7 = 58.5
- Total social cost: $\frac{4.3\times74.5+5.7\times58.5}{10} \approx 65.41$

Comparing to theoretical benchmarks (under asymmetric costs):

- Social optimum: $n_A \approx 5$, average cost ≈ 62.0 (minimum SC)
- Nash equilibrium: $n_A^* \approx 3.5$, average cost ≈ 62.5 (near-optimum)
- Independent forward-looking: converges close to Nash (≈ 62.5)
- Coalition-based: average cost ≈ 65.41 (~5% above optimum, ~4.6% above Nash)

Coalition formation achieves welfare close to the theoretical optimum and Nash levels, outperforming naive independent forward-looking learning in terms of average performance stability despite ongoing volatility. The small deviation from optimum reflects the tension between coordination gains and residual non-stationarity/deception effects.

4.2.3 Coalition Structure

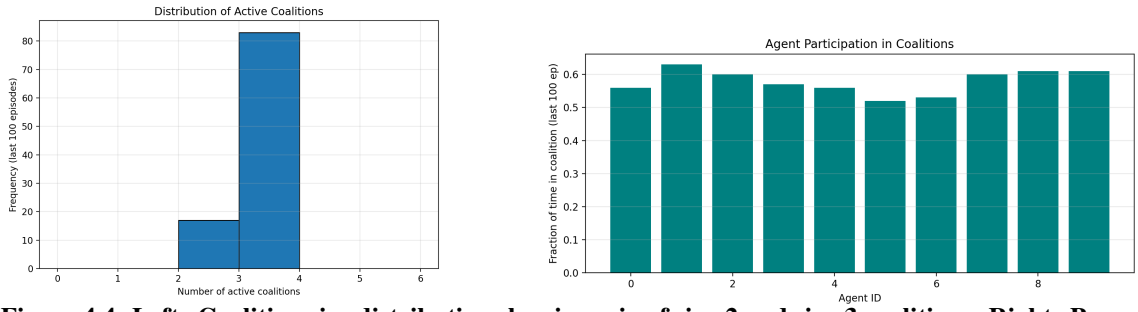

Figure 4.4. Left: Coalition size distribution showing mix of size-2 and size-3 coalitions. Right: Per-agent participation frequency revealing heterogeneous cooperation patterns.

Figure 4.4 reveals:

- **Size distribution:** Approximately equal frequency of 2-agent and 3-agent coalitions, with few singletons
- **Participation heterogeneity:** Some agents participate in 60-80% of episodes while others remain mostly independent

4.2.4 Coalition-Level Rewards

Final episode snapshot (Episode 3000):

- Coalition 1: Agents [1, 5], average reward = -60.0
- Coalition 2: Agents [6, 9], average reward = -65.0
- Coalition 3: Agent [0] alone (too small for analysis)

Aggregated statistics (last 100 episodes):

Table 4.3. Coalition reward statistics

Statistic	Value
Mean coalition reward	-71.64
Standard deviation	17.35
Minimum	-145.00
Maximum	-50.00

High variance (std = 17.35) indicates coalition rewards fluctuate substantially depending on route congestion and composition of other coalitions.

4.3 Game-Theoretic Metrics & Fairness

4.3.1 Shapley Value Comparison

Table 4.4. Learned imputation vs Shapley values (final episode)

Agent	Learned Imputation	Shapley Value	Difference
0	-65	-64	-1
1	-60	-64	+4
2	-65	-64	-1
3	-65	-64	-1
4	-55	-64	+9
5	-60	-64	+4
6	-65	-64	-1
7	-55	-64	+9
8	-65	-64	-1
9	-65	-64	-1
Sum	-620	-640	
Mean |diff|			**3.20**

Table 4.4 shows learned rewards deviate from Shapley values by mean absolute difference of 3.20. Notable patterns:

- Agents 4 and 7 receive substantially better allocations (-55 vs -64)
- Agents 1 and 5 also benefit (+4 advantage)
- Remaining agents receive approximately fair shares

Asymmetry suggests some agents extract more than fair share through favorable coalition matching or strategic defection. Equal-split allocation does not guarantee Shapley fairness when coalitions form opportunistically.

4.3.2 Core Stability Analysis

Core membership test: **Failed**

Violating coalition: Agent 9 acting alone achieves better reward than current allocation.

The learned coalition structure is unstable. No allocation exists that prevents all profitable deviations. This explains high volatility in Figure 4.3: agents continuously experiment with different partners because no stable configuration exists.

Empty core is expected for congestion games (Rosenthal, 1973): individual incentives conflict with coalition stability. Despite instability, time-averaged outcomes remain near-optimal (5.1 vs 5.0), suggesting volatility may prevent lock-in to inefficient structures.

DISCUSSION AND CONCLUSION

5.1 Insights

Coalition formation overcomes forward-looking learning failures. While independent agents with $\gamma = 0.9$ converge to inefficient Nash equilibrium, coalition-based learning with identical parameters achieves near-optimal outcomes. Cooperation provides structured coordination that individual forward-looking optimization cannot.

Instability does not prevent good time-averaged performance. Despite continuous coalition reformation and core violations, average outcomes approach social optimum. Volatility may actually be beneficial by preventing premature lock-in.

Myopic learning remains competitive. Coalition mechanisms add substantial complexity (coalition formation, defection detection, reward sharing, game-theoretic analysis) yet achieve similar welfare to simple myopic independent learning. This raises questions about cost-benefit tradeoffs in real implementations.

Free-riding persists. Average coalition usage of 74% indicates some agents remain independent while benefiting from reduced congestion created by coordinating coalitions. This free-rider problem prevents full cooperation.

5.2 Limitations

Parameter sensitivity: Coalition participation (30%) and defection (10%) probabilities held fixed. Different values may substantially alter dynamics.

Symmetric agents: All agents identical in capabilities and preferences. Heterogeneity may change coalition formation incentives.

Simple deception model: Defection is random rather than strategic. Sophisticated agents might learn when to defect based on expected gains.

Scalability: Shapley value calculation requires enumerating 2^N coalitions, limiting analysis to small populations. Approximation methods needed for large-scale applications.

No explicit communication: Coalition formation is implicit through reward sharing. Explicit communication protocols might improve coordination.

5.3 Future Directions

- Test coalition mechanisms at larger scales (N=50, 100) to assess scalability
- Implement strategic defection: agents learn when to defect based on coalition composition
- Compare alternative allocation rules (Shapley-based, Nash bargaining) to equal-split
- Allow agents to learn coalition participation probability as part of Q-learning
- Investigate explicit communication channels for coalition coordination
- Test in asymmetric congestion games (different origins/destinations, heterogeneous costs)

5.4 Summary

Coalition-based Q-learning achieves near-optimal welfare (average cost 5.1 vs optimum 5.0) in settings where independent forward-looking learning fails catastrophically (average cost 2.0 vs optimum 5.0). This 97

Surprisingly, simple myopic independent learning ($\gamma = 0$) achieves comparable welfare to complex coalition mechanisms, suggesting that reducing temporal reasoning may be more effective than introducing cooperation in highly non-stationary congestion games.

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